WEEK 9 ASSIGNMENT
H524 Introduction to Biostatistics - Fall 2025

Total Points: 10 — **Due:** Sunday 11:59 PM

Part A: Multiple Choice Questions (7 points)

Answer the following 7 questions on correlation and simple linear regression. Each question is worth 1 point.

Question 1: Correlation Coefficient Range (1 point)

The Pearson correlation coefficient (r) ranges from:

A. 0 to 1
B. -1 to 0
C. -1 to 1
D. $-\infty$ to $+\infty$

Question 2: Interpreting Correlation (1 point)

A correlation coefficient of r = -0.85 indicates:

A. A weak positive linear relationship
B. A strong positive linear relationship
C. A weak negative linear relationship
D. A strong negative linear relationship

Question 3: Correlation vs. Causation (1 point)

If two variables have a strong positive correlation (r = 0.90), this means:

A. One variable causes changes in the other
B. The variables tend to increase together, but this does not prove causation
C. There is no relationship between the variables
D. The regression line has a slope of 0.90

Question 4: Simple Linear Regression Equation (1 point)

In the regression equation $\hat{y} = a + bx$, what does "b" represent?

A. The y-intercept
B. The slope of the regression line
C. The correlation coefficient
D. The predicted value of y

Question 5: Coefficient of Determination (1 point)

If the correlation coefficient r = 0.60, what is the coefficient of determination (R^2)?

A. 0.36
B. 0.60
C. 0.77
D. 1.67

Question 6: Interpreting R^2 (1 point)

An R^2 value of 0.64 means:

A. 64% of the variance in y is explained by x
B. 64% of the variance in x is explained by y
C. The correlation is $r = 0.64$
D. The slope of the regression line is 0.64

Question 7: Regression Assumptions (1 point)

Which of the following is NOT an assumption of simple linear regression?

A. Linearity (relationship between x and y is linear)
B. Independence of observations
C. Normality of residuals
D. The independent variable (x) must be normally distributed

Part B: Group Project Check-In - Presentation Readiness (3 points)

GROUP PROJECT: FINAL CHECK BEFORE PRESENTATIONS

Reminder: Presentations are NEXT WEEK (Week 10)!

- **Monday 12-1:20pm:** Groups 1 & 2
- **Wednesday 12-1:20pm:** Groups 3 & 4

Your Assignment

Group Leader submits evidence that your presentation is ready:

Choose ONE of the following to submit:

Option A: Presentation Outline (preferred)

- List of slides/sections (bullet points)
- Who will present each section
- Timing estimate for each part

Option B: Draft Slide Deck

- PowerPoint, Google Slides, or PDF
- Doesn't need to be final, but should show structure

Option C: Presentation Plan

- Written description of what you'll cover
- Division of speaking roles
- Confirmation that all members will participate

Submission Format

Submit as: File upload (PDF, PPT, DOCX) OR text entry

Length:

- Outline: 1 page
- Slides: Any number (draft is fine)
- Plan: 1-2 paragraphs

Grading: 3 points for demonstrating presentation preparation

Presentation Reminders

Your presentation should include:

- Brief dataset overview
- Key statistical findings
- **AI comparison highlights** (30-40% of presentation!)
- Conclusions and what you learned

Requirements:

- 15 minutes (+ 5 min Q&A)
- All team members participate
- Use slides
- Attend your assigned day (Mon or Wed)

Time to finalize:

- This week: Draft slides and practice

- Next week: Final touches and delivery

Notes

- **Only the group leader submits**
- This check-in ensures you're ready for Week 10
- If you submit an outline now, you can refine slides this week
- Don't wait until the day before to start your presentation!
- **Grading (10% of course grade) is next week -** take this seriously!